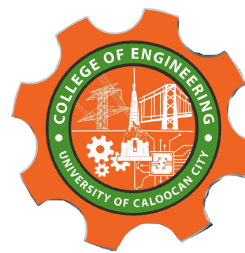




**UNIVERSITY OF CALOOCAN CITY
COMPUTER ENGINEERING DEPARTMENT**



Software Design

Progress Report No. 5

Requirement Analysis

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I. Objectives

In this section, the goals in this laboratory are:

- To illustrate the process of determining the requirements for a system
- To enumerate both the requirement definition and requirement specification

II. Methods

Requirement elicitation is one of the critical parts of the process. We must use various techniques to determine what the users need and what the customers really want.

Two Kinds of Requirements:

1. Requirement definition – a documentation that is written in terms that the customer can understand.
2. Requirement specification – a documentation that the definition are written in technical terms appropriate to the development of the entire system

Background

Over the semester you will be developing a “System”. This will be a suite of applications of real options methods to a professional issue of your own choosing. The object of developing this “System” is to provide you with a worked-out application of real options analysis in an area of your interest. The expectation is that, by creating this system, you will achieve an in-depth understanding of how real system analysis could be useful to you professionally.

This “System” might also provide you with a good start on some research project or thesis or be a helpful complement to the work you were doing – that would be an added benefit. In general, you are encouraged to develop this “System” around some research or course project that you might be involved with.

Laboratory Activity

1. Describe your System Application
2. Describe the engineering system that you would like to analyze this semester.
3. Choose something that interests you and that you wish to explore, as you will spend substantial time on it;
4. Choose any ways to gather requirements (survey, questionnaires, Use Case diagram, and the like)

You should place yourself in the position of someone participating in the design of this system. For example, you might want to put yourself in the role of a system designer.

****Be sure to describe the following:**

1. What is the system? What does it include, and what does it exclude?
2. What are its principal design levers or variables? What can we manipulate to improve its performance? What kind of model do we have to analyze the effect of these variables?
3. What are the benefits of this system? What models do we have of how the design affects these variables?

III. Results

Laboratory Activity

1. Describe your System Application

A salon management system is the system application that we are suggesting. The purpose of this program is to assist salons in managing and automating their everyday operations on a single platform. Online appointment scheduling, reminders, and post-visit feedback are all made possible. The system facilitates the management of scheduling, customer information, services provided, billing, inventory, and promotions for salon employees and owners. The technology enhances productivity, accuracy, and the general customer experience by minimizing manual labor and digitally organizing data.

2. Describe the engineering system that you would like to analyze this semester

This semester, we will examine an engineering system that is a software-based information system that automates salon business activities. The system consists of a number of parts, including staff scheduling, inventory tracking, invoicing and payment processing, customer relationship management (CRM), appointment scheduling, and marketing tools.

The system's processing and storage of inputs (customer information, reservations, services, and payments) and outputs (confirmations, receipts, reports, reminders, and feedback) will be examined from an engineering standpoint. The system's functionality, user interface, dependability, and how automation addresses typical salon issues like duplicate reservations, no-shows, improper data management, and ineffective processes will all be highlighted.

3. Choose something that interests you and that you wish to explore, as you will spend substantial time on it;

Our group decided to research a salon management system because, as the figure illustrates, it directly addresses actual issues that salon firms frequently encounter. Salons without automated systems frequently face problems like manual appointment scheduling,

no-shows, customer data confusion, invoicing inefficiencies, inventory shortages, and staff scheduling conflicts.

We are interested in this method because it demonstrates how automation may address real-world, everyday business issues. Technology may increase productivity and customer satisfaction with features like online booking, automated reminders, CRM systems, inventory tracking, and loyalty programs. This project enables us to investigate a practical system that we can evaluate and enhance over the course of the semester because salons are easily observed and frequently utilized.

4. Choose any ways to gather requirements (survey, questionnaires, Use Case diagram, and the like)

- **Surveys and Questionnaires**

Customers and employees of the salon will be given surveys to learn about typical problems like booking challenges, lengthy wait times, missing appointments, and feedback gathering. This makes it easier to choose which automatic features are most important.

- **Diagrams of use cases**

To depict interactions between users (customers, employees, and administrators) and the system, use case diagrams will be made. Appointment scheduling, reminders, inventory control, payment processing, and feedback gathering are a few examples.

- **Observation**

We can better identify workflow issues that automation can address, such as overbooked stylists, wasted staff time, and sluggish checkout procedures, by observing everyday salon operations.

- **Interviews**

Salon owners or managers will be interviewed about business issues such as staff scheduling, inventory control, billing problems, and customer retention tactics.

You should place yourself in the position of someone participating in the design of this system. For example, you might want to put yourself in the role of a system designer.

1. What is the system? What does it include, and what does it exclude?

The system is a salon management system made to automate and oversee a salon's everyday activities. Online appointment scheduling, customer data management, staff scheduling, billing and payment processing, inventory tracking, marketing tools, and feedback gathering are just a few of its capabilities. Customers, salon employees, and owners can communicate with the salon more effectively and efficiently thanks to the technology.

However, salon services like hair cutting, styling, and cosmetic treatments are not included in the system. Additionally, interior management, staff training procedures, and actual salon equipment are not included. The system's primary focus is on automation and information management rather than service delivery.

2. What are its principal design levers or variables? What can we manipulate to improve its performance? What kind of model do we have to analyze the effect of these variables?

The components that can be changed to enhance performance are the system's primary design levers or variables. Rules for scheduling appointments, frequency of reminders, staff availability, price and discount options, inventory restocking levels, and customer loyalty features are a few of these. Salon operators may lower no-shows, enhance staff productivity, effectively manage inventory, and boost client retention by modifying these factors.

We can use data flow diagrams, use case diagrams, and process flow models to examine the consequences of these variables. The effects of modifications to booking policies or reminder systems on wait times and no-show rates can also be investigated using basic simulation models. These models make it easier to see how the system functions in various scenarios.

3. What are the benefits of this system? What models do we have of how the design affects these variables?

The Salon Management System's primary advantages are enhanced productivity, fewer mistakes, better customer satisfaction, higher income, and better decision-making. Automation guarantees precise record keeping, minimizes scheduling conflicts, and cuts down on manual labor. Personalized services, reminders, and simpler booking are all beneficial to customers.

Performance models like service time analysis, customer retention models, and inventory consumption models can be used to examine the connection between system design and these advantages. For instance, CRM features are associated with increased customer loyalty, but automated reminders are associated with decreased no-show rates. These models aid in demonstrating how salon performance is directly enhanced by well-designed systems.

CONCLUSION

The Salon Management System project shows how technology may address typical salon issues like customer retention, scheduling, and record-keeping. In order to make sure the software satisfies actual user needs and enhances customer happiness and business efficiency, it offers practical experience in system analysis through surveys, interviews, and diagrams.

REFERENCE

1. [1] Co Arthur O.. “University of Caloocan City Computer Engineering Department Honor Code,” UCC-CpE Departmental Policies, 2020.